

DataSci 451 Final Project Report

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Project Overview

The goal of this project is to model and predict housing prices using a hierarchical Bayesian regression framework. The full dataset comes from a large USA real estate dataset published on Kaggle, but only a random sample of 5,000 observations was used due to the size of the original file. The model accounts for differences across both cities and states while using house size in square feet and acres as the main predictors. A baseline linear regression was first built for comparison, followed by a hierarchical model fit in Stan to allow regression relationships to vary by location. Model performance was evaluated using in-sample R^2 and posterior predictive distributions for selected cities.

Data Processing

The original Kaggle dataset was too large to be processed efficiently, so a random sample of 5,000 observations was taken for this project. Before sampling we cleaned the data set and removed outliers and rows with null values. We removed houses that were more than 10 million dollars, had over 50 acres and had more than 10,000 square feet. The data was sampled from the following five states: Massachusetts, California, Michigan, Texas and Florida. For each state we chose ten cities and for each city we randomly sampled 100 houses. The specific cities and states used for posterior prediction were chosen based on a mix of personal relevance and data availability. We selected California because Skye is from Los Angeles, and Massachusetts because the Myles is from Boston. We also included Ann Arbor, Michigan since it is where we currently attend university. The remaining cities and states were chosen based on having the highest number of observations in our dataset, ensuring more stable posterior estimates and clearer comparisons across locations. From this sample, price, acres, city, state, and house size were retained. All continuous variables and the response variable were standardized using z-scores to improve numerical stability and make coefficients comparable. The standardized formula we used was $Z_x = \frac{X - \bar{X}}{\text{sd}(\bar{X})}$. City and state were converted into factor variables, and unique ID values were created for each city and state to support hierarchical indexing in Stan. A design matrix was constructed using an intercept, standardized house

size, and standardized acre size, and all required components were assembled into a list for model fitting.

Exploratory Data Analysis

Below is the distribution of house prices in the sample.

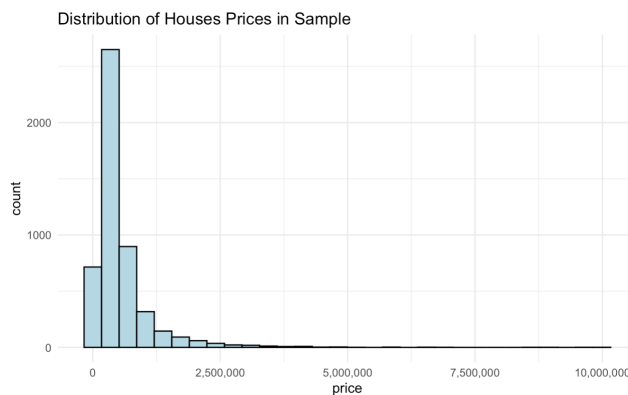


Figure 1: Distribution of house prices within the sample

The sampled data show that housing prices are heavily right-skewed, with a small number of extremely expensive homes relative to the median price.

Next we assessed the relationship between acres and house size to insure the variables were not correlated with each other. Below is the relationship between acres and log square footage

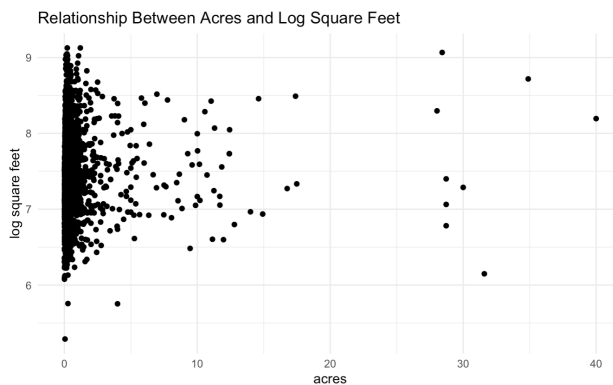


Figure 2: Relationship of log square feet and acres within the sample

From figure 2, we can see there is no correlation between acres and log square feet ensuring that autocorrelation will not be a problem we have to consider for modeling the house prices with the hierarchical regression model.

Model Description

In this section of the report, we explain how our hierarchical model was structured and designed. For our report we created two models. The first model was a simple multi regression model with no hierarchical structure. The second model is the hierarchical regression model. We used the first model to act as a baseline model to compare with the hierarchical regression model.

Baseline Model

For the baseline regression model, we did not consider the city and the state of the house. We only consider the covariates, square feet and acres. Below is a description of the baseline model, the covariates and response variable are standardized.:

$$\widehat{\text{Price}} = \beta_0 + \beta_{\text{sq. feet}} * \text{sq. feet} + \beta_{\text{acres}} * \text{acres}$$

The in sample R^2 for this model was 0.226 indicating that the model is not a good model for modeling house prices. The results of this model make sense as the price of a house is depended on where it is located not only its features.

Hierarchical Model

For the hierarchical regression model, we use a bayesian hierarchical structure based on city and state to sample the coefficients for the regression model. The hierarchical structure has three levels, the first level is the state level, the second level is the city level, and the third level is the data level. The hierarchical model samples a vector of the coefficients for each city and state, the covariants and response are standardized. Let $k = 1, 2, 3$ be the number of coefficient vectors for each city and state which include the intercept, square feet. Let $s = 1, \dots, 5$ be the number of states. Let $c = 1, \dots, 10$ be the number of cities per state. Let $i = 1, \dots, 100$ be the number of houses per city. Let $y_{i,s,c}$ be the house price and let $x_{i,s,c}$ be the house covariates.

Hyperprior:

For each covariate $k = 1, 2, 3$

$$\omega_k \sim N(0, 2)$$

State Level (Level 1):

For each state $s = 1, \dots, 5$

For each covariate $k = 1, 2, 3$

$$\beta_{s,k} \mid \omega_k \sim N(\omega_k, \sigma_{\text{state},k}^2)$$

$$\sigma_{\text{state},k}^2 \sim N(0, 1)$$

$$\text{with } \sigma_{\text{state},k}^2 \geq 0$$

City Level (Level 2):

For each city $c = 1, \dots, 10$ in state s

For each covariate $k = 1, 2, 3$

$$\beta_{s,c,k} \mid \beta_{s,k} \sim N(\beta_{s,k}, \sigma_{\text{city},k}^2)$$

$$\sigma_{\text{city},k}^2 \sim N(0, 1)$$

$$\text{with } \sigma_{\text{city},k}^2 \geq 0$$

Data Level (Level 3):

For each house $i = 1, \dots, 100$ in city c in state s

$$y_{i,s,c} \mid \vec{\beta}_{s,c} \sim N(\mu_{i,s,c}, \sigma^2)$$

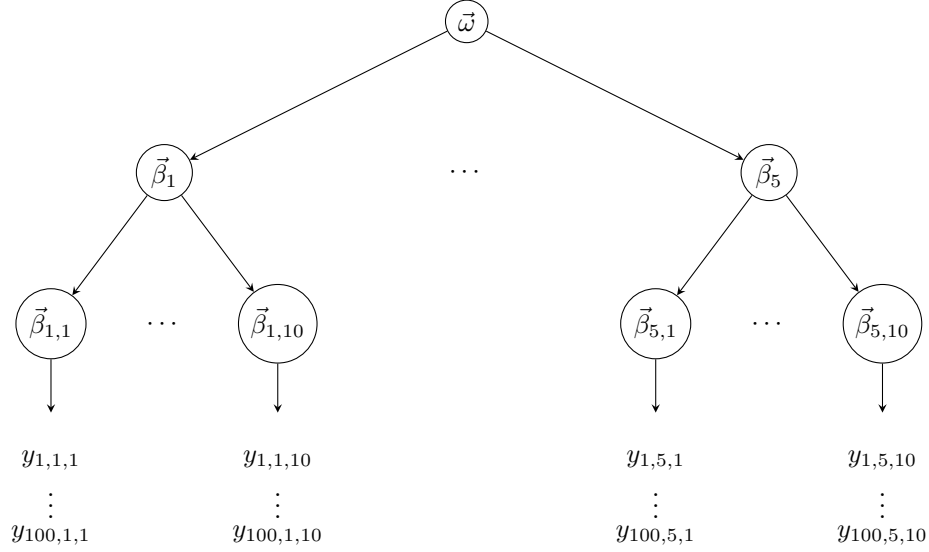
$$\sigma^2 \sim N(0, 1)$$

$$\text{with } \sigma^2 \geq 0$$

Where:

$$\mu_{i,s,c} = \vec{\beta}_{s,c}^T x_{i,s,c}$$

Hierarchical Model Diagram



STAN Sampling Algorithm

We used the STAN NUTS sampling algorithm to sample our parameters for the hierarchical model. The sampling algorithm included 4 independent markov chains with 1000 warmup iterations, 2000 sample iterations. We used a set random seed for consistent model reproducibility.

Model Results

To assess the quality of the hierarchical regression model we calculated the in sample R^2 using the mean posterior values for the city level coefficients. R^2 was 0.72 showing a drastic improvement over the baseline model and indicating that the hierarchical structure for the coefficients was effective for modeling house prices.

Trace Plots

Figure 3, Figure 4 and Figure 5 display the trace plots which visualize the markov chain sampling algorithm for some of our parameters. These trace plots show that the sampling algorithm converged effectively.

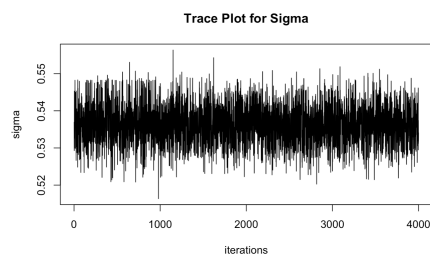


Figure 3: Trace plot for sigma

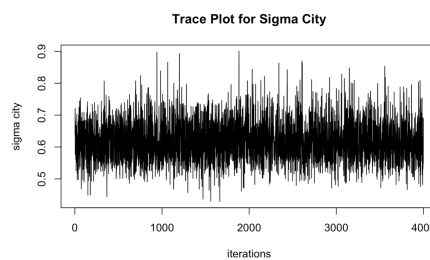


Figure 4: Trace plot for sigma city level

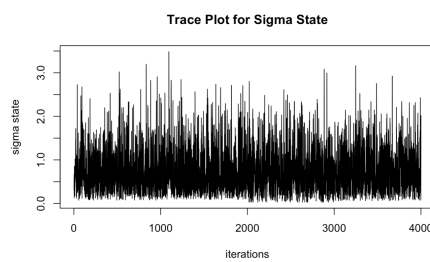


Figure 5: Trace plot for sigma state level

Effective Sample Sizes

We computed the effective sample size of all 175 parameters that were sampled in the hierarchical model. Below is a boxplot of the effective sample sizes for all parameters.

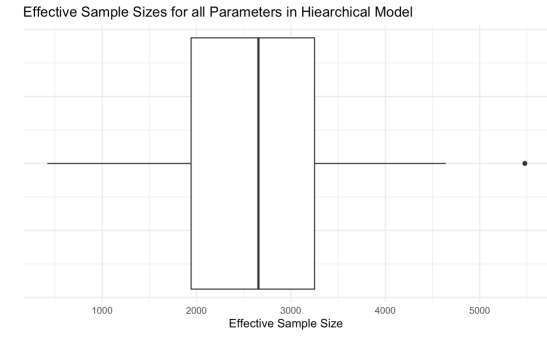


Figure 6: Box plot of effective sample sizes for all parameters

Figure 6 shows most parameters have an effective sample size over 1,000 with the inter quartile range effective sample size around 2,000 to 3,200. These effective sample size values indicate our markov chains mixed well with no autocorrelation. The sigma state parameters for each beta coefficient had lower effective sample sizes to rest the of the parameters with 418, 442, and 472. These effective sample sizes are lower because the parameters are scale parameters for the state level coefficients. Since the state coefficients shrink towards the global hyperparameters, the scale parameters for the coefficients have a smaller effective sample size.

Posterior Results

We computed the posterior price prediction distributions for different cities using the posterior city coefficients from the hierarchical model. We used the same house covariates for all the posterior price predictions across all the cities, 0.2 acres and 1600 square feet (roughly the median house features in the sample). Below is a plot that shows the 95% posterior price prediction confidence interval for the 0.2 acre 1600 square foot house for the following cities: Boston, San Francisco, Los Angeles, Detroit, Ann Arbor, Houston, Dallas, San Antonio, Miami and Tampa.

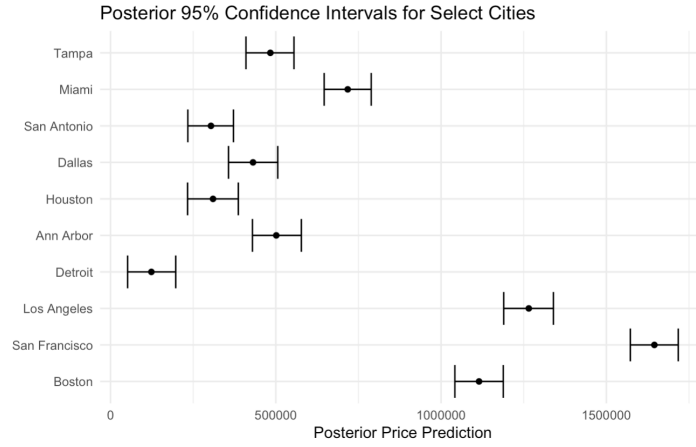


Figure 7: 95% posterior price prediction confidence intervals for different cities

Figure 7 shows that for the same house features, the posterior predicted price intervals are all different for each city. The plot shows that Boston, San Francisco, and Los Angeles have far more expensive intervals than the other cities with each interval exceed \$1,000,000 dollars. The other cities excluding Detroit have intervals in the range of \$250,000 dollars to \$750,000 dollars. Detroit's interval ranges from \$50,000 dollars to \$250,000 dollars.

References

Sakib, A. S. (2024, March 30). *USA Real Estate Dataset*. Kaggle. <https://www.kaggle.com/datasets/ahmedshahriarsakib/usa-real-estate-dataset/data>

Statement of Contribution: Myles

For this project I went to GSI office hours with my partner to discuss a outline structure for the hierarchical model. Leading up to the project presentation, I tested and tweaked the hierarchical model to optimize the best results for the project. I wrote the official model in with professional mathematical syntax. I helped find the data and come up with a plan to sample and preprocess the data. I came up with the idea of implementing a baseline regression model to compare with the hierarchical model. I extracted the results of the final hierarchical model including the visuals of the MCMC and the posterior results. Together we split the work between creating the final presentation and writing this report.

Statement of Contribution: Skye

I came up with the original idea for the project by combining both my partner's interests and mine into a single focused topic. We worked together in office hours to develop the foundational structure of the hierarchical model and clarify our overall approach. My partner handled more of the programming than I did, while I focused on shaping the direction of the project. We jointly created the presentation slides and collaborated equally on writing this report.